

Sessional Exam – I
Subject: Formal Languages and Automata Theory
Paper Code: BCE-C612

Time: 1 Hr

MM: 20

Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

Q.No.	Section A (Questions of 6 marks each)	CO																								
Q1.	Identify the differences between DFA & NFA or Describe a Finite Automata with e-moves.	CO1																								
Q2.	Define Finite Automaton with the help of diagram. Or Describe the transition system	CO1																								
	Section B (Questions of 8 marks each)																									
Q3.	Compare the Mealy and Moore Machine and convert the following Moore machine to equivalent Mealy Machine <table><tr><th>Present State</th><th colspan="2">Next State</th><th>Output</th></tr><tr><td></td><th>a=0</th><th>a=1</th><td></td></tr><tr><td>->q0</td><td>q3</td><td>q1</td><td>0</td></tr><tr><td>q1</td><td>q1</td><td>q2</td><td>1</td></tr><tr><td>q2</td><td>q2</td><td>q3</td><td>0</td></tr><tr><td>q3</td><td>q3</td><td>q3</td><td>0</td></tr></table> Or Explain Grammer with its definition and If $G = (\{S\}, \{0,1\}, \{S \rightarrow 0S1, S \rightarrow \Lambda\}, S)$, find $L(G)$	Present State	Next State		Output		a=0	a=1		->q0	q3	q1	0	q1	q1	q2	1	q2	q2	q3	0	q3	q3	q3	0	CO2
Present State	Next State		Output																							
	a=0	a=1																								
->q0	q3	q1	0																							
q1	q1	q2	1																							
q2	q2	q3	0																							
q3	q3	q3	0																							

Second Sessional Exam
Subject: Formal Languages and Automata Theory
Paper Code: BCE-C612

Time: 1 Hr

MM: 20

Q.No.	Section A (Questions of 6 marks each)	CO
Q1.	What is a pushdown automata? Explain Or What is a Turing machine? Explain	CO3
Q2.	Construct a pda A accepting $L = \{wcw^T \mid w \in \{a,b\}^*\}$ by final state. Or Describe the different types of Turing Machines.	CO3
	Section B (Questions of 8 marks each)	
Q3.	Construct a pda A accepting the set of all strings over $\{a, b\}$ with equal number of a's and b's. or M is a Turing machine represented by the transition system in Fig. given below. Obtain the computation sequence of M for processing the input string 0011.	CO4

